# Time of Season and Game Segment Is Not Related to Likelihood of Lower-Limb Injuries: A Meta-Analysis

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**Abstract**

**Objective:** Fatigue due to game play is often cited as a factor in musculoskeletal injuries; however, it is unclear whether or not this view is supported by published research findings. Given the importance researchers and practitioners place on the potential effects of game play with respect to injury, it is important to understand what inferences can be drawn from the collective research in this realm. This meta-analysis will consider the time of season and segment of the game, as it relates anterior cruciate ligament (ACL), groin, and hamstring injury occurrence. **Data Sources:** Database searches were run in PubMed, MEDLINE, SportDiscus, CINAHL, and Ausport, in addition to the inclusion of articles identified manually. **Study Selection:** Search terms were chosen to identify articles related to each of the 3 injuries of interest. There were no date limitations placed on the articles, as such, all published articles listed in the databases up to November 2017 were eligible for selection if they met the search criteria. **Data Extraction:** Initial searches yielded 1349 articles, and this was eventually reduced to 15 articles deemed suitable for inclusion in the meta-analysis, which provided 21 data sets. **Data Synthesis:** Comparing the first half with the second half of the season, there were no differences in ACL, groin, or hamstring injury occurrences [ACL: odds ratio (OR), 1.27; confidence interval (CI), 0.43-3.78, groin: OR, 1.79; CI, 0.63-5.06, and hamstring: OR, 1.16; CI, 0.88-1.53]. Similarly, there were no differences in injury occurrence between the first and second halves of the game for the ACL or hamstring injuries (ACL: OR, 0.43; CI, 0.47-7.92, hamstring: OR, 0.85; CI, 0.58-1.24). **Conclusions:** Findings from this meta-analysis determined that time in season or time in game does not influence risk of the ACL, groin, or hamstring injury. Commonly, many studies did not provide sufficient detail to be included in the meta-analysis. Consequently, it is recommended that future studies report data related to the timing of the injury within the season or game.

**Key Words:** injury prevention, sports injury, epidemiology

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## INTRODUCTION

In certain cases, musculoskeletal injuries in sport can be a devastating event for athletes and, although not always, may be season- and/or career-ending.[1] There is a large body of literature that seeks to identify risk factors in order that coaches and athletes can target these risks in a hope to limit the injuries sustained by athletes.[2–4] Of particular interest are lower-limb injuries, including those associated with the knee, ankle, hip, and posterior thigh, because of their potential severity with respect to an athlete's career.[5] Common risk factors, such as age, sex, and previous injury, for example, have been well researched.[4,6,7] Although researchers have attempted to investigate, what is often referred to as, fatigue and its relation to physical performance and injury prediction, the literature is ambiguous with respect to the findings.[8,9]

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As summarized by Barber-Westin and Noyes,[9] it is often hypothesized that fatigued muscles absorb less energy leading to an overstretching of structures, which in turn reach their limit and fail. In practice, it is difficult, if not impossible, to measure this directly, and instead, researchers have sought to quantify the manifestation of fatigue using measures such as high-speed running and "training load."[10,11] Essentially, these articles investigate an athlete's ability to tolerate "training load"[11] or sustain an intensity of running speeds.[10] Although these may be useful measures, or indicators, of fatigue and can be used to monitor an athlete, they do not inform about the absolute consequence of fatigue, which is injury.

In Australia, two-thirds of injuries in the Australian Football League are related to the knee, ankle, or hip.[12] Although the relative contribution of these injuries to the total number of injuries has decreased, the absolute number is largely unchanged.[12] In the National Football League in the United States, anterior cruciate ligament (ACL) injuries are an area of concern. Although there was a recent drop in the absolute number of ACL tears (49 in 2014), that number has grown over the last 2 seasons reaching 56 in 2016.[13] Interestingly, the total number of injuries has not decreased, despite an increased understanding of the mechanisms of injuries and evidence-based intervention strategies.[14]

Several studies successfully decreased musculoskeletal injuries by inclusion of targeted injury prevention programs with decreases of over 60% in some cases.[2,15,16] However, these studies have not considered the time of the season or game time in their programs, ie, they do not train athletes when they may have a residual level of fatigue. Currently,



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there is evidence that lower-limb mechanics change due to laboratory-induced acute fatigue.[17–19] These studies, and other similar studies, are limited, as there are no direct relationships with the occurrence of an injury. As such, these studies only confirm that the movement patterns are altered as a result of the fatiguing protocol. Furthermore, in a recent review of laboratory-based fatigue protocols on lower-limb function concluded that fatigue protocols do not act to change biomechanical factors that lead to increased ACL injury risk.[9] When considering fatigue due to actual game play, there is limited research that has quantified the effect of time of season or game segment on injury occurrence, which may potentially assist practitioners with further targeted injury prevention programs to a time of season, or level of fatigue within a game.

Recently, Okoroha et al[20] concluded that the minutes played in a basketball game was not a predictor of ACL injury. On the other hand, other researchers have found that the time of the match does influence when injuries occur.[21,22] Although these studies refer to different injuries (ACL tears vs hamstring strains), the findings are seemingly at odds with one another. Therefore, the aim of this meta-analysis was to determine whether injuries are clustered within a particular point in the season or segment of the game.

## METHODS

This meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[23]

### *Search Strategy*

Three searches, 1 per injury type, were run in the following 5 databases: PubMed, MEDLINE, SportDiscus, CINAHL, and Ausport. The search terms and operators are detailed in Table 1, with each injury type run as a separate search. Keywords within a category were included in the search using "OR," and each category was included using "AND" (Table 1). There were no date limitations placed on the articles, as such, all published articles listed in the databases up to November 2017 were eligible for selection if they met the search criteria. Finally, manual searches of the databases and literature were conducted to supplement automated database searches.

### *Selection Criteria*

To be deemed suitable for inclusion, articles needed to be written in English, be original research, and contain timing of injury occurrence with respect to the season or game segment. Therefore, the aim of this meta-analysis was to determine whether time of the season or game segment had any effect on sustaining a lower-limb injury. Only injuries sustained to the ACL, groin, or hamstring were considered for this research. Finally, only level 1 sports, as defined by Daniel et al,[24] were included (eg, basketball, football, etc.).

Two reviewers independently applied the selection criteria to the studies. The reviewers first screened titles and abstracts, and those articles that clearly did not meet the inclusion criteria were excluded. Full-text versions of remaining studies were obtained, and criteria again applied independently by the 2 reviewers. Where reviewers disagreed on the inclusion or exclusion of articles during their independent reviews, they collectively assessed the article to make a joint decision about its inclusion or exclusion. If further dispute between the reviewers ensued, a third independent reviewer assessed and made the final decision. Two articles warranted follow-up with the original authors to seek original data for inclusion in the analysis. One author no longer had access to the data, and the second article's author did not respond to the repeated requests.

**TABLE 1. Search Strategy and Terms Used to Identify Potentially Suitable Articles**

| | ACL | Groin | Hamstring |
|---|---|---|---|
| Study type | epidemiolog* | epidemiolog* | epidemiolog* |
| Injury location | acl | groin | hamstring |
| | anterior cruciate ligament | adductor | posterior thigh |
| | | | semimembranosus |
| | | | semitendinosus |
| Sports | soccer | soccer | soccer |
| | basketball | basketball | basketball |
| | football | football | football |
| | rugby* | rugby* | rugby* |
| | sport* | sport* | sport* |
| Injury type | injur* | injur* | injur* |
| | ruptur* | ruptur* | ruptur* |
| | | | strain |
| | | | tear* |
| | | | pain* |
| | | | dysfunction |
| | | | trauma* |

*Each of the injury areas (columns) was run as separate searches, terms within a cell were included in the search using "OR," and each cell was included in the search using "AND."*
** Asterisk indicates wildcard for search terms.*
*ACL, anterior cruciate ligament.*



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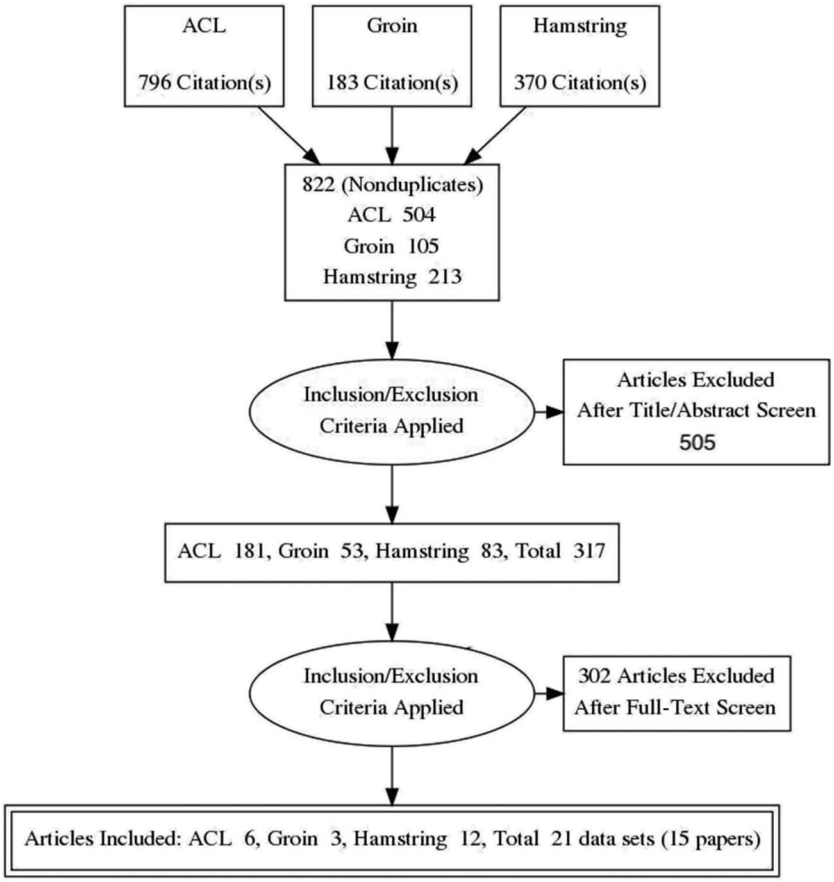


**Figure 1.** PRISMA flow diagram showing movement of articles through the selection process. Note that 4 articles contained data relating to game and season injuries, resulting in 21 sets of data.

### *Data Extraction and Synthesis*

Two reviewers independently extracted data from each study. The data extracted included the number of injuries, timing (segment of season or match) of the injury occurrence, and the total sample size during the study period (injuries sustained during training were not included). Participant details (age and sex) and sport were also recorded. Where studies provided figures of results and not raw numbers, reviewers used customized MATLAB software (GRABIT, v1.0.0.1) to digitize the figure and extract the raw injury data. In addition, if data were presented as percentages or relative occurrences, raw injury numbers were calculated. The timing of injury occurrence was recorded either as segment of the season (first or second half) or game segment (by half or quarter). Injuries recorded during the season were recorded as having occurred in either the first or second half of the season. Some studies provided injury data per month, not by season halves; in these cases, the data were split evenly to include the same number of months in the first half of the season as the second half. Where there was an odd number of months, the number of injuries in the middle month was divided by 2 and added to the season half totals. For injuries that occurred during a match, the quarter within which the injury occurred was recorded. Odds ratio (OR) meta-analyses were conducted to evaluate the time of injury occurrence. For each of the target injuries, pooled estimates were summarized in forest plots, presented separately for season (first compared with second half) and match (first half vs second half, first vs second quarter, first quarter vs fourth quarter, or second to third quarters). Owing to the heterogeneity of the study samples, a random effects model was used. The degree of heterogeneity within the results of included studies was assessed using the I2 index, with larger scores indicating that a greater proportion of the variability in the results could be attributed to heterogeneity.

## RESULTS

The flow of journal articles through the selection process is contained in Figure 1.

### *Study Selection*

The initial 3 searches, 1 for each injury region, yielded 1349 articles, and this was reduced to 822 after duplicates were removed. Further screening of the abstracts removed 505 articles, and the full text of these remaining 317 articles was gathered for additional scrutiny. Ultimately, 21 total sets of data from 15 articles were deemed suitable for inclusion in the meta-analysis, ie, they contained data specific to the number of injuries occurring to the ACL, groin, or hamstring, and data regarding the timing of the injury within a season or game. For the groin, there was only 1 study with suitable data for game segment analysis, as such, a meta-analysis was only run for the groin looking at the effect of season.



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**TABLE 2. Summary of Studies Used for Injury Data Across a Season**

| Author | Sport | Injury Location | First-Half Season | Second-Half Season |
|---|---|---|---|---|
| Dallalana et al[26] | Rugby union | ACL | 5 | 2 |
| Dodson et al[28] | American football | ACL | 77 | 53 |
| Orchard[29] | Australian rules football | ACL | 31 | 43 |
| Hägglund et al[7] | Soccer | Groin | 247 | 240 |
| Jackson et al[30] | Basketball | Groin | 282 | 135 |
| Dalton et al[31] | Various NCAA sports | Hamstring | 301 | 211 |
| Ekstrand et al[33] | Soccer | Hamstring | 485 | 556 |
| Hägglund et al[7] | Soccer | Hamstring | 421 | 435 |
| Jackson et al[30] | Basketball | Hamstring | 257 | 160 |
| Roe et al[32] | Gaelic football | Hamstring | 172 | 153 |
| Timmins et al[34] | Soccer | Hamstring | 8 | 9 |
| Woods et al[21] | Soccer | Hamstring | 318 | 318 |

*ACL, anterior cruciate ligament.*

### *Risk of Bias Assessment*

The 2 reviewers who selected the articles independently assessed the quality of each article using the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies developed by the National Heart, Lung, and Blood Institute.[25] Any discrepancies were resolved through consensus meetings between the 2 reviewers. The risk of bias tool used is not designed to provide a quantitative assessment of quality of the articles but to provide a means to assess the internal validity of the article through a series of 14 questions. Ultimately, only 1 question, "12. Were the outcome assessors blinded to the exposure status of participants?" scored lowly for all articles. Question 13, "Was loss to follow-up after baseline 20% or less?" was considered not relevant and, as such, not included when assessing study quality and risk of bias. Ultimately, 1 article was rated highly for 9 of 14 questions, all other articles rated highly for 11 or 12 questions. Consequently, the risk of bias was considered low for the articles included in the meta-analysis.

### *Effect of Season*

A comparison of the first with the second half of the season did not reveal any differences in occurrence of injury for ACL [OR, 1.27; confidence interval (CI), 0.43-3.78], groin (OR, 1.79; CI, 0.63-5.06), or hamstring (OR, 1.16; CI, 0.88-1.53). Note that an OR greater than 1 indicates that injuries tended to occur in the first half of the season. Provided in Table 2 are the number of injuries used from each study in the meta-analysis. The meta-analysis showed that when results from all studies were combined, there was no effect of season time on injury occurrence (Figures 2-4).

### *Effect of Game Segment*

Comparing the first half with the second half of a game, there were no differences in injury occurrence for the ACL (OR, 0.43; CI, 0.07-2.56) or hamstring (OR, 0.85; CI, 0.58-1.24). Similarly, comparing the first with the fourth quarter, there was no effect of game time for either the ACL (OR, 0.84; CI, 0.22-2.68) or hamstring (OR, 0.62; CI, 0.34-1.16). This was

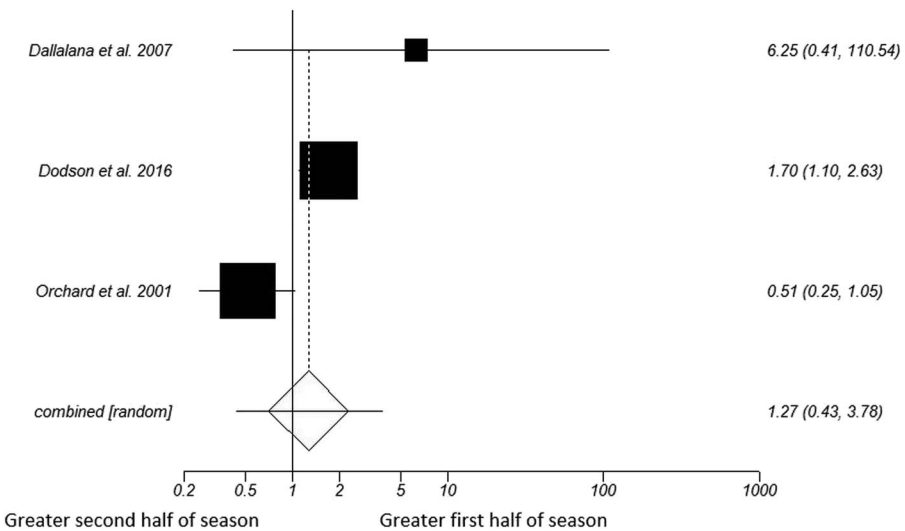


**Figure 2.** Forest plot of results from ACL injury articles looking at the first half of the season compared with the second half of the season. An OR of greater than 1 indicates more injuries in the first half of the season.



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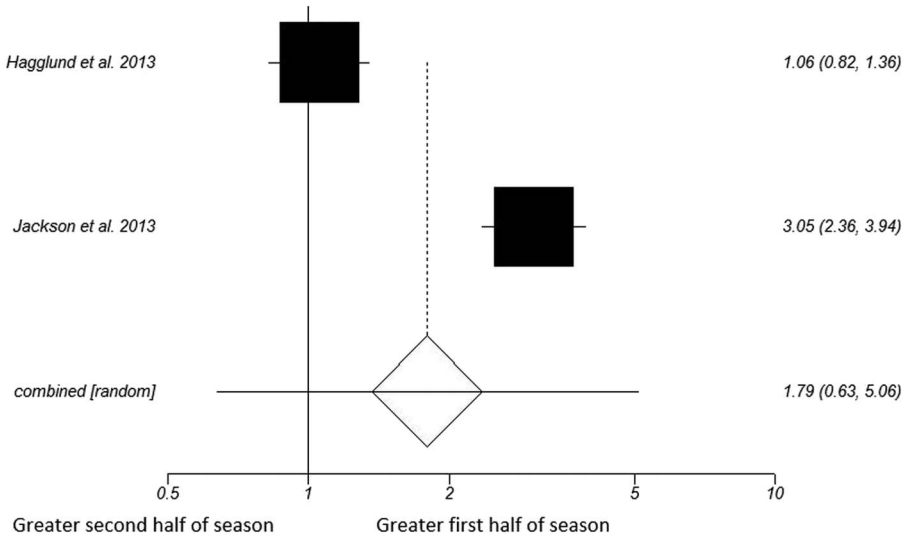


**Figure 3.** Forest plot of results from groin injury articles looking at the first half of the season compared with the second half of the season. An OR of greater than 1 indicates more injuries in the first half of the season.

the same for all game segment comparisons performed, ie, first to second, first to fourth, and second to third quarters. Note that an OR greater than 1 indicates that injuries tended to occur in the earlier game segment, eg, the first half/quarter compared with the second half/quarter. Provided in Table 3 are the number of injuries used, by game half, from each study in the meta-analysis. Table 4 details the ORs and CIs for all comparisons. Provided in Figures 5 and 6 are the results of the meta-analyses, which show that injuries did not tend to occur in particular game segments.

## DISCUSSION

The purpose of this meta-analysis was to determine whether lower-limb injuries, namely injuries to the ACL, groin, or hamstring, tend to cluster at particular times of the season or game. To that end, this meta-analysis determined that injuries do not tend to occur in either the first or second half of the season, nor do they tend to occur during a particular segment of the game. Consequently, the data denote that time of season does not influence the likelihood of sustaining an injury to the ACL, groin, or hamstring. Likewise, game play duration is not related to sustaining an injury to the ACL or hamstring. Specifically, whether or not this is related to fatigue cannot be stated as the data analyzed in this current meta-analysis looked simply at the timing of injury, within a season and game, and not fatigue measures. Noting that it would be extremely difficult to measure fatigue, by whatever metric may be preferred, at the time of an injury.

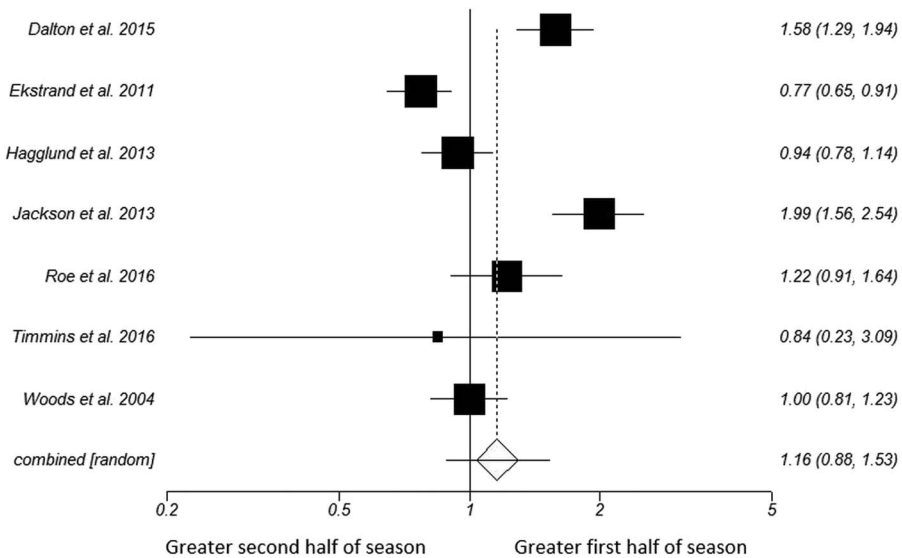


**Figure 4.** Forest plot of results from hamstring injury articles looking at the first half of the season compared with the second half of the season. An OR of greater than 1 indicates more injuries in the first half of the season.



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**TABLE 3. Summary of Studies Used for Injury Data Within a Game**

| Author | Sport | Injury Location | First-Half Game | Second-Half Game |
|---|---|---|---|---|
| Dallalana et al[26] | Rugby union | ACL | 2 | 5 |
| Okoroha et al[20] | Basketball | ACL | 18 | 24 |
| Pangrazio and Forriol[35] | Soccer | ACL | 1 | 0 |
| Ekstrand et al[33],* | Soccer | Groin | 58 | 68 |
| Brooks et al[27] | Rugby union | Hamstring | 66 | 98 |
| Cross et al[22] | Soccer | Hamstring | 292 | 228 |
| Ekstrand et al[37] | Soccer | Hamstring | 121 | 121 |
| Roe et al[32] | Gaelic football | Hamstring | 168 | 199 |
| Woods et al[21] | Soccer | Hamstring | 346 | 403 |

** Data are not used for meta-analysis, as only 1 data set available.*
*ACL, anterior cruciate ligament.*

Although over 1000 articles were originally identified as being potentially suitable, many did not provide the level of detail with respect to the timing of injuries (ie, when the injury occurred within the season or game, or a suitable breakdown of injury types). Future research should report a greater level of detail of injuries than currently tends to be reported in the literature; this will enable an improved understanding of injury characteristics in future meta-analyses. As a result of the limited availability of injury statistics related time of season/game or injury location, effect of game segment could only be assessed for ACL and hamstring injuries.

It must also be noted that in the current meta-analysis, all sports were considered together, as were both males and females. However, there are different patterns of injuries seen for different sports, eg, Dallalana et al[26] and Brooks et al[27] suggest that in rugby union, ACL and hamstring injuries, respectively, are related to game time; however, Okoroha et al[20] contradicts this finding in National Basketball Association (NBA). Likewise, Cross et al[22] found evidence that there are sex differences when looking at hamstring injuries in soccer. Consequently, researchers should aim to investigate sport and sex influences on injuries with respect to time in the season and game segment.

Also, worth noting is that this meta-analysis looked at absolute numbers of injuries, ie, not relative to player-exposures, minutes played, or any other normalizing metric. The intent of this meta-analysis was to look at a big picture of when injuries occur within a season or game. Understanding this may help practitioners to focus specific injury prevention programs, or strategies, where a pattern of injury occurrence exists. However, when looking across all sports, there does not appear to be any such pattern. Noting the comments above suggesting there may be sport- and sex-specific influences on injury distributions.

**TABLE 4. Summary of OR and CIs for All Comparisons Made**

| Injury | OR | CI |
|---|---|---|
| First- vs second-half season | | |
| ACL | 1.27 | 0.43-3.78 |
| Groin | 1.79 | 0.63-5.06 |
| Hamstring | 1.16 | 0.88-1.53 |
| First- vs second-half game | | |
| ACL | 0.43 | 0.47-7.92 |
| Hamstring | 0.85 | 0.58-1.24 |
| First- vs second-quarter game | | |
| ACL | 1.92 | 0.47-7.92 |
| Hamstring | 0.77 | 0.56-1.04 |
| First- vs third-quarter game | | |
| ACL | 0.59 | 0.10-3.62 |
| Hamstring | 1.03 | 0.82-1.28 |
| First- vs fourth-quarter game | | |
| ACL | 0.84 | 0.27-2.68 |
| Hamstring | 0.62 | 0.34-1.16 |
| Second- vs third-quarter game | | |
| ACL | 0.36 | 0.10-1.34 |
| Hamstring | 1.33 | 0.92-1.91 |

*ACL, anterior cruciate ligament; CI, confidence interval, OR, odds ratio.*



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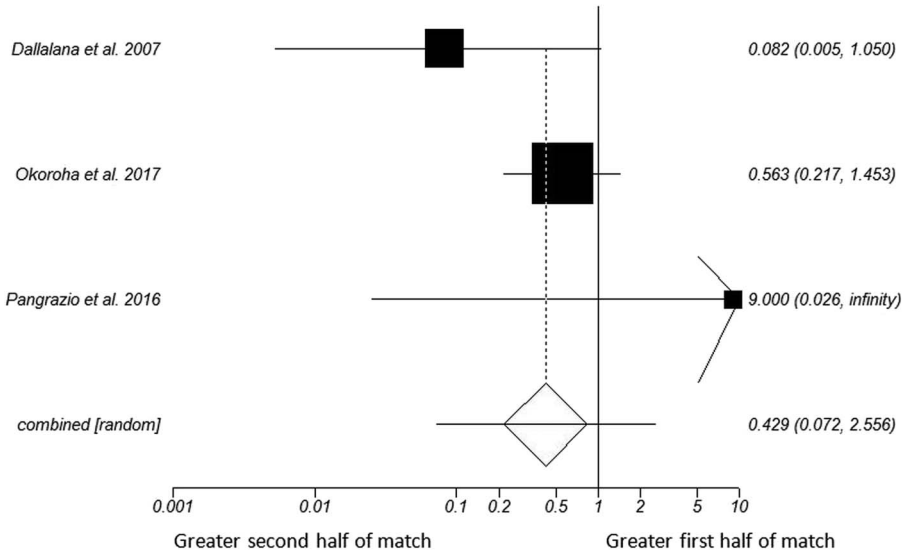


**Figure 5.** Forest plot of results from ACL injury articles looking at the first half of the match compared with the second half of the match. An OR of greater than 1 indicates more injuries in the first half of the match.

### *Effect of Season*

Results from the meta-analysis determined that for ACL, groin, and hamstring injuries, there was no effect of season on when injuries tended to occur; results from individual studies, however, are mixed. For ACL injuries, mostly consistent findings within all 3 studies suggested injuries are more likely to occur early, not late, in the season.[26,28,29] Dallalana et al[26] and Dodson et al[28] both concluded this based on total number of injuries; however, when factors such as player numbers were considered, they concluded that the differences were not meaningful. Orchard,[29] however, made this claim based on relative injuries, although a greater number of ACL injuries occurred late in the season, compared with the early season.

Both Jackson et al[30] and Hägglund et al[7] investigated groin injuries with largely similar findings to hamstring injuries. Jackson et al[30] reported 693 groin injuries in the first half of the season compared with 394 in the second half of the season. Although Hägglund et al[7] report more groin injuries in the first half of the season compared with the second, the difference is quite minor, ie, 247 injuries versus 240 injuries. Neither research group however looked specifically at time in the regular season and the relationship with injury, instead only comparing the preseason with the entire competitive season.

Findings from hamstring studies were considerably more mixed than the ACL findings. Of the 7 studies included, 3 reported an increased number of injuries early in the

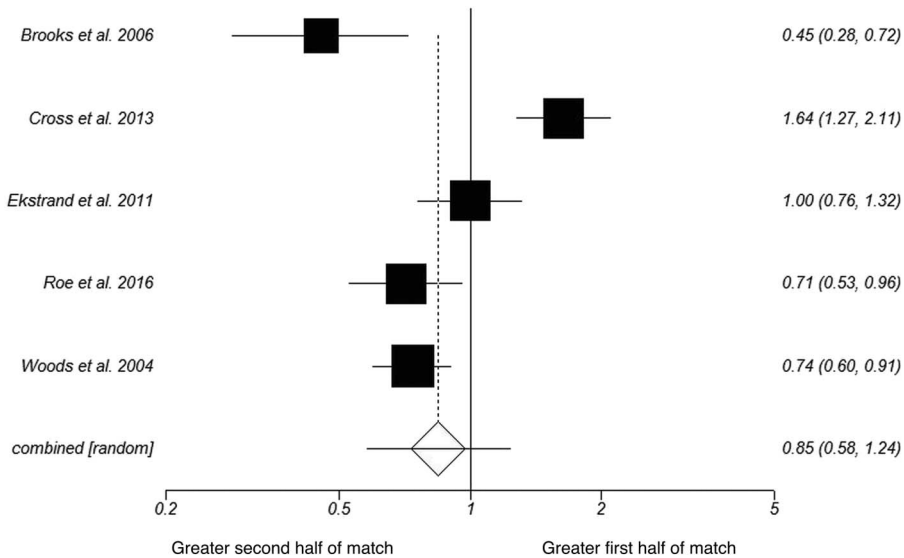


**Figure 6.** Forest plot of results from hamstring injury articles looking at the first half of the match compared with the second half of the match. An OR of greater than 1 indicates more injuries in the first half of the match.



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season,[30–32] 3 reported the second half of the season as being more injurious,[7,33,34] and 1 study reported the same number of injuries in each season half.[21] It is unsurprising then that a clear finding toward the first or second half of the season was not found in the current meta-analysis. However, none of the studies concluded time of season, excluding preseason, to be related to injury likelihood, or no comment was made by the researchers. Of particular note, Dalton et al[31] investigated over 1000 injuries in 25 sports and in their assessment of all injuries as a whole, and, by individual sport, concluded that time in season did not affect the hamstring injury rate.

### *Effect of Segment (Half/Quarter) of Game*

The results from this meta-analysis support recent research by Okoroha et al[20] who concluded that the number of minutes played does not relate to ACL injuries in NBA athletes. Their research reported 24 injuries in the second half of the NBA season, compared with 18 in the first half. However, breaking the game down even further, the majority of injuries, 13, occurred in the first quarter, although the differences in injury rates were not significant. These results differ from those of Dallalana et al[26] who investigated English rugby union knee injuries, and they found the majority of all 177 knee injuries they recorded during matches occurred in the second half of the game, approximately 60%. The authors went even further and found that the first quarter of rugby union matches had the least amount of injuries. For the purposes of this meta-analysis however, only ACL ruptures were considered, which included 5 (72%) in the second half and 2 (28%) in the first. This pattern matches that seen for all 211 knee injuries they observed during matches and training. Finally, Pangrazio and Forriol,[35] although included in the meta-analysis, only reported a single ACL injury that occurred in the first half. It is possible the difference between the NBA and rugby union studies is due to the different sports; however, with only 1 study identified for each sport, a definitive determination cannot be made.

Hamstring injury studies similarly were mixed, which is supported by the current meta-analysis (Figure 4). Two of the studies reported more injuries in the second half compared with the first,[27,32] 1 study found the first half to be more injurious,[22] and another reported the same number of injuries in each half.[33] Brooks et al,[27] in investigating hamstring injuries in rugby union, concluded that forwards were more likely to be injured the longer the game continued, whereas backs were more likely to be injured in the latter part of each half of the game. Although Cross et al[22] reported more injuries in the first than the second half when looking at males and females as 1 cohort, they concluded that females, in fact, sustain more injuries in the second half of a soccer game for first time hamstring injuries (62%), and this value jumps even higher (76%) for recurrent injuries.[36] Similar to ACL injuries, it is once again possible that the different sporting codes have different injury patterns; it seems also that injury patterns, with respect to game time, may differ between sexes.

## CONCLUSIONS

From the studies available currently, it appears that injuries do not cluster whether it be related to the time of the season or time of the game when considering ACL, groin, or hamstring injuries. Noting that some individual studies, as discussed above, suggested there may be sport- and sex-specific trends. Despite the large number of potentially useful research studies that were identified for all injuries, surprisingly few contained sufficient detail regarding the timing of the injury occurrence within the season and/or game. It is recommended that future studies should report data related to the timing of the injury within the season or game, as appropriate, to allow researchers and practitioners to gain a further understanding of how injuries are, or are not, affected by fatigue related to the season or game.

## References


1. Hamrin Senorski E, Seil R, Svantesson E, et al. "I never made it to the pros…" Return to sport and becoming an elite athlete after pediatric and adolescent anterior cruciate ligament injury—current evidence and future directions. *Knee Surg Sports Traumatol Arthrosc*. 2018;26:1011–1018.
2. Finch CF, Twomey DM, Fortington LV, et al. Preventing Australian football injuries with a targeted neuromuscular control exercise programme: comparative injury rates from a training intervention delivered in a clustered randomised controlled trial. *Inj Prev*. 2016;22: 123–128.
3. Silvers-Granelli HJ, Bizzini M, Arundale A, et al. Does the FIFA 11+ injury prevention program reduce the incidence of ACL injury in male soccer players? *Clin Orthop Relat Res*. 2017;475:2447–2455.
4. Zebis MK, Andersen LL, Brandt M, et al. Effects of evidence-based prevention training on neuromuscular and biomechanical risk factors for ACL injury in adolescent female athletes: a randomised controlled trial. *Br J Sports Med*. 2016;50:552–557.
5. Weber CD, Horst K, Nguyen AR, et al. Return to sports after multiple trauma: which factors are responsible?-results from a 17-year follow-up. *Clin J Sport Med*. 2017;27:481–486.
6. Emery CA, Meeuwisse WH. Risk factors for groin injuries in hockey. *Med Sci Sports Exerc*. 2001;33:1423–1433.
7. Hägglund M, Waldén M, Ekstrand J. Risk factors for lower extremity muscle injury in professional soccer: the UEFA Injury Study. *Am J Sports Med*. 2013;41:327–335.
8. Cairns SP, Knicker AJ, Thompson MW, et al. Evaluation of models used to study neuromuscular fatigue. *Exerc Sport Sci Rev*. 2005;33:9–16.
9. Barber-Westin SD, Noyes FR. Effect of fatigue protocols on lower limb neuromuscular function and implications for anterior cruciate ligament injury prevention training: a systematic review. *Am J Sports Med*. 2017; 45:3388–3396.
10. Coutts AJ, Quinn J, Hocking J, et al. Match running performance in elite Australian Rules Football. *J Sci Med Sport*. 2010;13:543–548.
11. Gabbett TJ. The development and application of an injury prediction model for noncontact, soft-tissue injuries in elite collision sport athletes. *J Strength Cond Res*. 2010;24:2593–2603.
12. Australian Football League. *2016 AFL Injury Survey*. Melbourne, Victoria, Australia: Australian Football League; 2016.
13. National Football League. *NFL Health and Safety Report*. New York, New York: United States National Football League; 2017.
14. Shield AJ, Bourne MN. Hamstring injury prevention practices in elite sport: evidence for eccentric strength vs. lumbo-pelvic training. *Sports Med*. 2018;48:513–524.
15. Sadigursky D, Braid JA, De Lira DNL, et al. The FIFA 11 + injury prevention program for soccer players: a systematic review. *BMC Sports Sci Med Rehabil*. 2017;9:18.
16. Hewett TE, Lindenfeld TN, Riccobene JV, et al. The effect of neuromuscular training on the incidence of knee injury in female athletes. A prospective study. *Am J Sports Med*. 1999;27:699–706.
17. Cortes N, Greska E, Kollock R, et al. Changes in lower extremity biomechanics due to a short-term fatigue protocol. *J Athl Train*. 2013;48:306–313.
18. Savage RJ, Lay BS, Wills JA, et al. Prolonged running increases knee moments in sidestepping and cutting manoeuvres in sport. *J Sci Med Sport*. 2018;21:508–512.
19. Sanna G, O'Connor KM. Fatigue-related changes in stance leg mechanics during sidestep cutting maneuvers. *Clin Biomech (Bristol, Avon)*. 2008; 23:946–954.
20. Okoroha KR, Marfo K, Meta F, et al. Amount of minutes played does not contribute to anterior cruciate ligament injury in National Basketball Association athletes. *Orthopedics*. 2017;40:e658–e662.




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21. Woods C, Hawkins RD, Maltby S, et al. The Football Association Medical Research Programme: an audit of injuries in professional football—analysis of hamstring injuries. *Br J Sports Med*. 2004;38:36–41.
22. Cross KM, Gurka KK, Saliba S, et al. Comparison of hamstring strain injury rates between male and female intercollegiate soccer athletes. *Am J Sports Med*. 2013;41:742–748.
23. Liberati A, Altman DG, Tetzlaff J, et al. The PRISMA statement for reporting systematic reviews and meta-analyses of studies that evaluate healthcare interventions: explanation and elaboration. *BMJ*. 2009;339:b2700.
24. Daniel DM, Stone ML, Dobson BE, et al. Fate of the ACL-injured patient: A prospective outcome study. *Am J Sports Med*. 1994;22:632–644.
25. *Study Quality Assessment Tools*. 2018. Available at: https://www.nhlbi.nih.gov/health-topics/study-quality-assessment-tools. Accessed March 15, 2018.
26. Dallalana RJ, Brooks JH, Kemp SP, et al. The epidemiology of knee injuries in English professional rugby union. *Am J Sports Med*. 2007;35:818–830.
27. Brooks JH, Fuller CW, Kemp SP, et al. Incidence, risk, and prevention of hamstring muscle injuries in professional rugby union. *Am J Sports Med*. 2006;34:1297–1306.
28. Dodson CC, Secrist ES, Bhat SB, et al. Anterior cruciate ligament injuries in National Football League athletes from 2010 to 2013. *Orthop J Sports Med*. 2016;4:1.
29. Orchard J. The AFL penetrometer study: work in progress. *J Sci Med Sport*. 2001;4:220–232.
30. Jackson TJ, Starkey C, McElhiney D, et al. Epidemiology of hip injuries in the National Basketball Association: a 24-year overview. *Orthop J Sports Med*. 2013;1:2325967113499130.
31. Dalton SL, Kerr ZY, Dompier TP. Epidemiology of hamstring strains in 25 NCAA sports in the 2009-2010 to 2013-2014 academic years. *Am J Sports Med*. 2015;43:2671–2679.
32. Roe M, Murphy JC, Gissane C, et al. Hamstring injuries in elite Gaelic football: an 8-year investigation to identify injury rates, time-loss patterns and players at increased risk. *Br J Sports Med*. 2016;52:982–988.
33. Ekstrand J, Hägglund M, Waldén M, et al. Injury incidence and injury patterns in professional football: the UEFA injury study. *Br J Sports Med*. 2011;45:553–558.
34. Timmins RG, Bourne MN, Shield AJ, et al. Short biceps femoris fascicles and eccentric knee flexor weakness increase the risk of hamstring injury in elite football (soccer): a prospective cohort study. *Br J Sports Med*. 2016;50:1524–1535.
35. Pangrazio O, Forriol F. Epidemiology of soccer players traumatic injuries during the 2015 America Cup. *Muscles Ligaments Tendons J*. 2016;6:124–130.
36. Paterno MV, Rauh MJ, Schmitt LC, et al. Incidence of second ACL injuries 2 years after primary ACL reconstruction and return to sport. *Am J Sports Med*. 2014;42:1567–1573.
37. Ekstrand J, Hägglund M, Waldén M. Epidemiology of muscle injuries in professional football (soccer). *Am J Sports Med*. 2011;39:1226–1232.




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